

Jungseob Lee

NLP Researcher · Ph.D. Student, Korea University

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Summary

Integrated M.S./Ph.D. researcher at Korea University NLP Lab (Advisor: Prof. Heuiseok Lim), working on large language models and natural language processing, with a focus on agentic LLMs, LLM efficiency, and efficient reasoning. First-author work at **EMNLP 2026** (Main), **ACL 2025** (Oral), and **Findings of ACL 2024**, with four papers accepted at **EMNLP 2026** (2 Main, 2 Findings). Served as post-training Project Manager (Korea University representative) for **VAETKI**, Korea's sovereign foundation model.

Research Interests: Large Language Models, Agentic LLM, LLM Efficiency, Efficient Reasoning

Education

Integrated M.S./Ph.D. in Computer Science 2021 – Present
Korea University, Seoul, Korea **Expected: Feb. 2027**
NLP Lab, Advisor: Prof. Heuiseok Lim

Experience

VAETKI, Korean Sovereign AI Foundation Model Jun 2025 – Jan 2026
Post-training Project Manager, Korea University Representative
Led post-training (SFT, RL, dataset curation) for Korea University in the NC AI (NCSOFT)-led 14-institution consortium of the Ministry of Science and ICT's national program. Open-sourced in Dec 2025, state of the art on CLIcK, KoBALT, KMMLU-Pro, and IFEval. [Model] [Technical Report]

Polyglot, Open-Source Multilingual Language Models 2022
Open-Source Contributor, EleutherAI
Worked on efficient language transfer of pretrained language models with cross-lingual post-training (XPT) and MAD-X adapters, and also took part in pretraining. [GitHub]

Awards & Honors

Outstanding Reviewer Award 2026
The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), San Diego

Best Paper Award (Top 1) 2025
Annual Conference on Human and Cognitive Language Technology (HCLT 2025)
For *KULLM-RAG: A RAG-Specialized Large Language Model with Intelligent Self-Verification* (first author)

Minister of Science and ICT Award 2023
ICT Challenge 2023, Ministry of Science and ICT (organized by IITP)
For *KoCheckGPT*, Korean-specialized detection of LLM-generated documents, built with M. Kang and S. Lee and selected from 79 teams across 28 universities

Selected Publications

Underlined = me, * equal contribution. Full list (43 papers) on Google Scholar.

EMNLP 2026 Main
The Hallucination Signal Is a Mean Shift: Why Simple Probes Suffice
J. Lee, J. Seo, H. Lim [arXiv]

EMNLP 2026 Main
Language Chain in Alignment: Cross-Lingual Ranking Preference Optimization
S. Lee, M. Kim, **J. Lee**, H. Lim [arXiv] [Code]

EMNLP 2026 Findings
DART: Draft-Agreement Routing for Training-Free Adaptive Thinking Budgets in Hybrid Reasoning Models
J. Lee, S. Hong, S. Lee, *et al.* [arXiv] [Code]

EMNLP 2026 Findings
ThinkFuse: Trajectory-Aware Test-Time Fusion for Small Reasoning Models
M. Kang*, **J. Lee***, J. Seo, H. Lim [arXiv] [Code]

ICLR 2026 Poster
Improving Semantic Proximity in Information Retrieval through Cross-Lingual Alignment
S. Hong, Y. Jang, **J. Lee**, H. Moon, H. Lim [PDF]

KBS 2026 Journal
Towards Enhancing Natural Language Inference for Identifying Erroneous Outputs in Data-to-Text Generation
C. Ok, E.K. Lee, H. Ju, **J. Lee**, D. Oh *Knowledge-Based Systems* [DOI]

AACL 2026 Findings
Skin-Deep: A Geometric Diagnostic for Alignment Fragility in Large Language Model Representations
D.J. Lee*, **J. Lee***, S. Lee, *et al.* [arXiv] [Code]

ACL 2025 Oral
Cross-lingual Optimization for Language Transfer in Large Language Models
J. Lee, S. Hong, H. Moon, H.S. Lim [PDF]

ESWA 2025 Journal
An Analysis on Language Transfer of Pre-trained Language Model with Cross-lingual Post-training
S. Son*, C. Park*, **J. Lee***, *et al.* *Expert Systems with Applications* [DOI]

ACL 2024 Findings
Length-aware Byte Pair Encoding for Mitigating Over-segmentation in Korean Machine Translation
J. Lee, H. Moon, S. Lee, *et al.* [PDF]

Appl. Sci. 2024 Journal
Analysis of the Effectiveness of Model, Data, and User-Centric Approaches for Chat Application: A Case Study of BlenderBot 2.0
C. Park*, **J. Lee***, S. Son, *et al.* *Applied Sciences* [DOI]

Math. 2023 Journal
A Survey on Evaluation Metrics for Machine Translation
S. Lee*, **J. Lee***, H. Moon, *et al.* *Mathematics* (142 citations) [PDF]

ACL 2023 Demo
PEEP-Talk: A Situational Dialogue-based Chatbot for English Education
S. Lee, Y. Jang, C. Park, **J. Lee**, *et al.* [PDF]

Under Review

arXiv Oct 2026
Predicting and Repairing Merge Collapse in Large Language Models
J. Lee, S. Lee, S. Eo, *et al.* [arXiv] [Code]

arXiv Oct 2026
Distilling Directional Verification
J. Lee, S. Eo, S. Hong, *et al.* [arXiv] [Code]

arXiv Oct 2026
CAST: Cost-Aware Speculative Trees from One-Pass Block Drafters
J. Lee, S. Eo [arXiv] [Code]

arXiv Oct 2026
Refusal Localizes, the Damage Relocates: Safety Layers Under Few-Sample Fine-Tuning
J. Lee*, D.J. Lee*, S. Eo, *et al.* [arXiv] [Code]

arXiv Sep 2026
Recovering Off-Policy Supervision for Speculative Decoding
J. Lee, C. Park, S. Eo, H. Moon [arXiv] [Code]

arXiv Sep 2026
Risk-Controlled Selective LLM Answering by Pricing Label-Free Checks
D.J. Lee*, **J. Lee***, C. Park, H. Moon, H. Lim [arXiv] [Code]

arXiv Sep 2026
Where Activation Sparsity and KV-Cache Sparsity Cross in LLM Decoding
J. Lee, S. Lee, S. Hong, S. Eo, H. Lim [arXiv] [Code]

arXiv Sep 2026
Faster Block-Diffusion Serving with Distribution-Free Risk Guarantees
J. Lee, D.J. Lee, C. Park, S. Eo, H. Lim [arXiv] [Code]

arXiv Sep 2026
Vision Is Not Overhead: One-Pass Block Drafting for Lossless Speculative Decoding in Vision-Language Models
J. Lee, S. Hong, D.J. Lee, *et al.* [arXiv] [Code]

arXiv Jul 2026
Answer-Conditioned Chains of Thought Degrade Verifiable-Reasoning Distillation in Large Language Models
J. Lee, S. Lee, S. Son, *et al.* [arXiv] [Code]

arXiv Jun 2026
To Isolate or to Score? Model-Adaptive Assessment for Cost-Efficient Multi-Agent RAG
J. Lee, C. Park, H. Lim [arXiv] [Code]

arXiv Jun 2026
Beyond Penalizing Mistakes: Stabilizing Efficiency Training in Large Reasoning Models via Adaptive Correct-Only Rewards
J. Lee*, S. Lee*, S. Hong, *et al.* [arXiv] [Code]

arXiv Sep 2026
RenderRank: Learning to Rerank Text with Compressed Visual Tokens
S. Hong, Y. Jang, **J. Lee**, H. Moon, H. Lim [arXiv]

arXiv Jul 2026
LAMAR: An Open Language-Aware Multilingual Alignment Reranker
S. Hong, Y. Jang, **J. Lee**, S. Lee, H. Lim [arXiv]

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